MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 8: Modified Equation Analysis: Numerical Dissipation and Dispersion

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1 Introduction

In Lecture 7, we established that numerical stability requires $|G(\xi)| \leq 1$ for all resolvable Fourier phase angles $\xi = k\Delta x \in [-\pi, \pi]$. Under this criterion, the First-Order Upwind (FOU), Lax-Friedrichs (LxF), and Lax-Wendroff (LW) schemes are all stable when the CFL condition $|\nu| = \frac{|a|\Delta t}{\Delta x} \leq 1$ is satisfied.

However, stability alone does not tell the full story of how a scheme behaves in practice. When we simulate waves on a computer, two striking artifacts appear:

- **Numerical Dissipation (Diffusion):** First-order schemes damp out peak amplitudes and severely smear sharp fronts over time.
- **Numerical Dispersion:** Higher-order schemes preserve peak amplitudes better, but generate spurious post-shock or pre-shock wiggles (oscillations) near steep gradients.

In this lecture, we introduce **Modified Equation Analysis**, a powerful mathematical tool that reveals the exact continuous partial differential equation our discrete algebraic algorithm is *actually* solving. We will quantify both numerical dissipation (amplitude error) and numerical dispersion (phase velocity error), compare classical schemes, and state **Godunov's Theorem**.

2 Modified Equation Analysis

When deriving a finite difference method via Taylor series, we truncate higher-order terms. **Modified equation analysis** reverses this perspective: we substitute Taylor series expansions of all discrete terms back into the difference scheme, express higher time derivatives in terms of spatial derivatives using the differential equation itself, and determine the effective PDE satisfied by the numerical solution up to higher order.

2.1 Modified Equation for the First-Order Upwind Scheme

Recall the First-Order Upwind scheme for $a > 0$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \tag{1}$$

Expanding $u_j^{n+1} = u(x_j, t^n + \Delta t)$ and $u_{j-1}^n = u(x_j - \Delta x, t^n)$ in Taylor series:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} + \dots \tag{2}$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6}\frac{\partial^3 u}{\partial x^3} - \dots \tag{3}$$

Substituting equations (2) and (3) into (1):

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{a\Delta x}{2}\frac{\partial^2 u}{\partial x^2} - \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} - \frac{a\Delta x^2}{6}\frac{\partial^3 u}{\partial x^3} + \dots \tag{4}$$

2.1.1 Eliminating Time Derivatives via the Cauchy-Kovalevskaya Procedure

To express the right-hand side of equation (4) purely in space, we differentiate the leading-order equation $\frac{\partial u}{\partial t} = -a\frac{\partial u}{\partial x} + \mathcal{O}(\Delta t, \Delta x)$:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(-a\frac{\partial u}{\partial x}\right) = -a\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) = -a\frac{\partial}{\partial x}\left(-a\frac{\partial u}{\partial x}\right) = a^2\frac{\partial^2 u}{\partial x^2} + \mathcal{O}(\Delta t, \Delta x) \tag{5}$$

$$\frac{\partial^3 u}{\partial t^3} = -a^3\frac{\partial^3 u}{\partial x^3} + \mathcal{O}(\Delta t, \Delta x) \tag{6}$$

Substituting these into equation (4) and grouping powers:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \left(\frac{a\Delta x}{2} - \frac{a^2\Delta t}{2}\right)\frac{\partial^2 u}{\partial x^2} - \left(\frac{a\Delta x^2}{6} - \frac{a^3\Delta t^2}{6}\right)\frac{\partial^3 u}{\partial x^3} + \dots \tag{7}$$

Factoring out the CFL number $\nu = \frac{a\Delta t}{\Delta x}$:

$$\boxed{\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \underbrace{\frac{a\Delta x}{2}(1-\nu)}_{\text{Numerical Dissipation } \alpha_{\text{num}}}\frac{\partial^2 u}{\partial x^2} - \underbrace{\frac{a\Delta x^2}{6}(1-\nu^2)}_{\text{Numerical Dispersion } \beta_{\text{num}}}\frac{\partial^3 u}{\partial x^3} + \dots} \tag{8}$$

Physical Insights from the Modified Equation

- **Even spatial derivative ($\frac{\partial^2 u}{\partial x^2}$):** Acts like physical viscosity or heat conduction. When $\nu < 1$, the coefficient $\alpha_{\text{num}} = \frac{a\Delta x}{2}(1-\nu) > 0$, causing wave amplitudes to decay and sharp fronts to smear.
- **Odd spatial derivative ($\frac{\partial^3 u}{\partial x^3}$):** Acts like a dispersive wave term (similar to the Korteweg-de Vries equation for water waves). It causes different Fourier frequencies to travel at different speeds.
- **Magic CFL number ($\nu = 1$):** When $\nu = 1$, all error coefficients $\alpha_{\text{num}} = \beta_{\text{num}} = 0$ vanish! The numerical solution shifts exactly along the characteristic: $u_j^{n+1} = u_{j-1}^n$.

2.2 Modified Equation for the Lax-Wendroff Scheme

The Lax-Wendroff scheme was deliberately designed to eliminate the second-order diffusion error ($\mathcal{O}(\Delta x)$). Following the exact same expansion procedure for equation (25) of Lecture 7, we find that the first non-vanishing error term is of third order:

$$\boxed{\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\underbrace{\frac{a\Delta x^2}{6}(1-\nu^2)}_{\text{Leading Dispersive Error}}\frac{\partial^3 u}{\partial x^3} - \underbrace{\frac{a^2\Delta t\Delta x^2}{8}(1-\nu^2)}_{\text{Fourth-Order Dissipation}}\frac{\partial^4 u}{\partial x^4} + \dots} \tag{9}$$

Key distinction: Lax-Wendroff has **zero second-order artificial viscosity** ($\alpha_{\text{num}} = 0$). Its leading error is purely **dispersive** ($\sim \frac{\partial^3 u}{\partial x^3}$). This explains why Lax-Wendroff does not smear smooth waves, but produces oscillations near discontinuities!

3 Fourier Analysis: Dissipation vs. Dispersion

Writing the complex amplification factor in polar form:

$$G(\xi) = |G(\xi)|\, e^{-\mathrm{i}\phi(\xi)} \tag{10}$$

where:

- $|G(\xi)|$ is the **numerical amplitude ratio** per time step.
- $\phi(\xi) = -\arg(G(\xi))$ is the **numerical phase shift** per time step.

3.1 Phase Speed and Dispersion Relation

In the exact advection equation, every Fourier mode $e^{ik(x-at)}$ travels at the exact speed a. The exact phase shift across a time step Δt is:

$$\phi_{\text{exact}} = ka\Delta t = k\Delta x \left(\frac{a\Delta t}{\Delta x}\right) = \nu\,\xi \tag{11}$$

For a numerical scheme, the effective **numerical phase velocity** $c_{\text{num}}(\xi)$ is:

$$c_{\text{num}}(\xi) = \frac{\phi(\xi)}{k\,\Delta t} = a\left(\frac{\phi(\xi)}{\nu\,\xi}\right) \tag{12}$$

- If $\frac{c_{\text{num}}(\xi)}{a} = 1$: All frequencies travel at the true physical wave speed (no dispersion).
- If $\frac{c_{\text{num}}(\xi)}{a} < 1$: High-frequency wave components lag behind the main wave, creating **trailing (post-shock) oscillations**.
- If $\frac{c_{\text{num}}(\xi)}{a} > 1$: High-frequency components outrun the main wave, creating **leading (pre-shock) oscillations**.

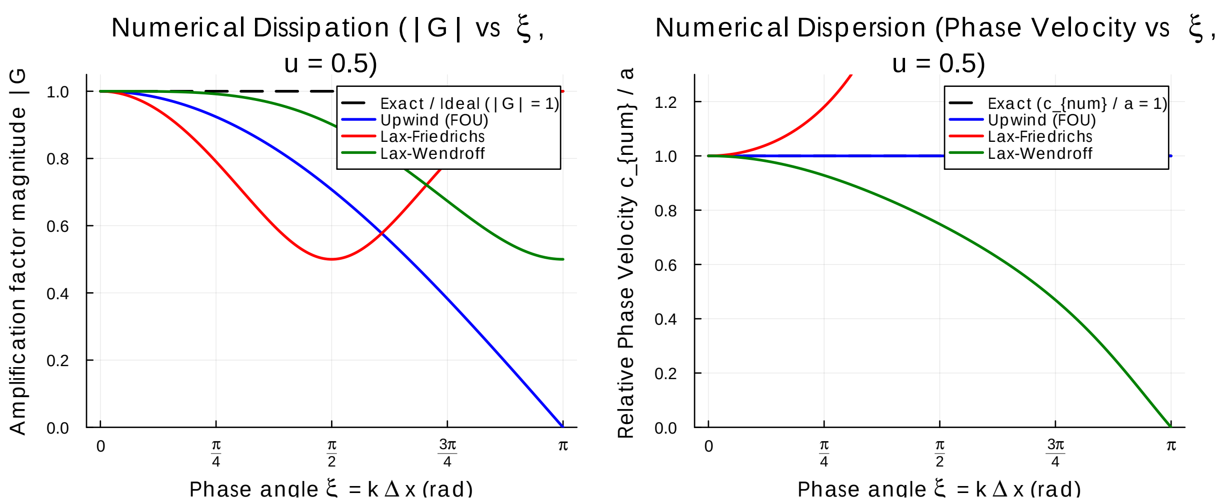

Figure 1: Left: Numerical dissipation ($|G(\xi)|$ vs phase angle $\xi = k\Delta x$ for CFL $\nu = 0.5$). Right: Numerical dispersion (Relative phase velocity c_{num}/a vs ξ). Generated with Julia.

4 Simulation Results: Smooth Waves vs. Discontinuous Fronts

To visualize the practical consequences of dissipation and dispersion, we simulate both a smooth Gaussian pulse and a discontinuous square wave under periodic boundary conditions using Julia.

From Figure 2, we observe:

1. **Upwind (FOU):** Monotonic (no oscillations), but broadens the pulse significantly due to $\mathcal{O}(\Delta x)$ numerical diffusion.
2. **Lax-Friedrichs:** Massive numerical diffusion ($\sim \frac{\Delta x^2}{2\Delta t}$); the square wave is completely degraded into a blurred hump.
3. **Lax-Wendroff:** High resolution and sharp gradient, but prominent **Gibbs-like oscillations** trail behind the discontinuous jumps. Crucially, the solution undershoots ($u < 0$) and overshoots ($u > 1$), creating non-physical extrema!

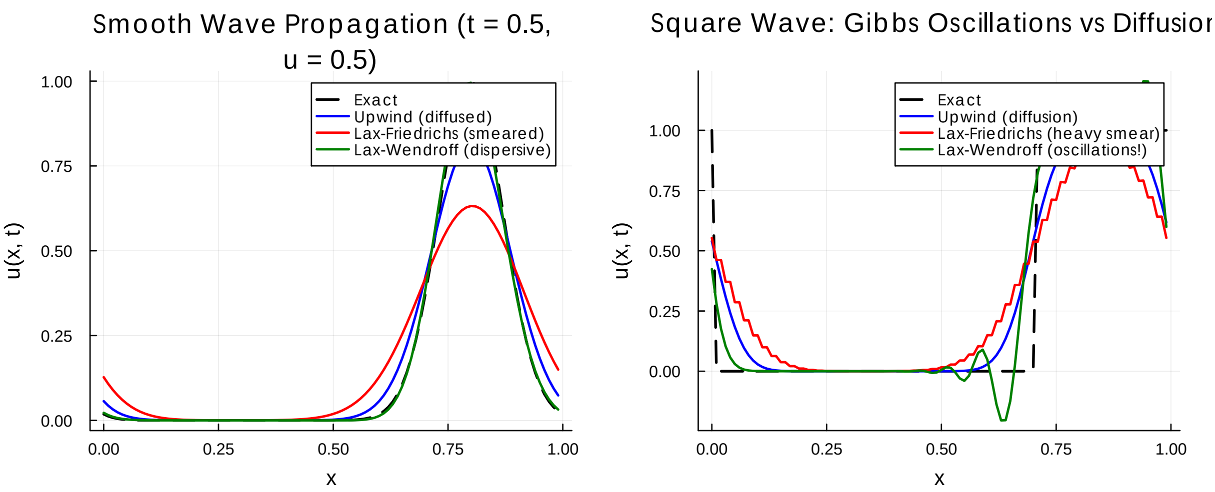

Figure 2: Advection simulation ($a = 1.0, \nu = 0.5$, grid $N = 100$). Left: Smooth Gaussian wave after half transit. Right: Discontinuous square wave showing heavy diffusion (Upwind, Lax-Friedrichs) vs dispersive post-shock oscillations (Lax-Wendroff). Generated with Julia.

5 Godunov's Barrier Theorem

This dilemma between excessive smearing (first-order schemes) and unphysical oscillations (second-order schemes) is not accidental. In 1959, Sergei Godunov proved a landmark impossibility result in numerical analysis:

Godunov's Theorem (1959)

Linear numerical schemes for solving partial differential equations cannot be simultaneously higher than first-order accurate and monotonicity-preserving (oscillation-free).

Implication for CFD: If we restrict ourselves to linear schemes with constant coefficients, we are forced to choose between the heavy blur of first-order schemes or the spurious oscillations of second-order schemes. To overcome this fundamental barrier, modern CFD relies on **non-linear high-resolution methods** (such as TVD schemes, flux limiters, and ENO/WENO methods, which we will study in Module 4).

6 Exercises

Exercises for Lecture 8

E1. Modified Equation for Lax-Friedrichs: Starting from the Lax-Friedrichs difference equation:

$$\frac{u_j^{n+1} - \frac{1}{2}(u_{j+1}^n + u_{j-1}^n)}{\Delta t} + a\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

use Taylor expansions and the Cauchy-Kovalevskaya procedure to derive its modified equation up to second order:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \frac{\Delta x^2}{2\Delta t}(1 - \nu^2)\frac{\partial^2 u}{\partial x^2}$$

Explain why Lax-Friedrichs is far more diffusive than Upwind when $\nu \ll 1$.

E2. Phase Velocity of Lax-Wendroff: For the Lax-Wendroff scheme, show that for small phase angles $\xi \ll 1$, the relative numerical phase velocity satisfies:

$$\frac{c_{\text{num}}(\xi)}{a} \approx 1 - \frac{1 - \nu^2}{6}\xi^2$$

Deduce that for any stable CFL number $|\nu| < 1$, high-frequency modes lag behind the physical wave ($c_{\text{num}} < a$), explaining why the oscillations in Figure 2 appear **behind** (to the left of) the moving front.

E3. Modified Equation for Diffusion Equation: For the heat equation $u_t = \alpha u_{xx}$ discretized by FTCS, derive the modified equation up to fourth spatial order and identify the leading numerical error term.

References

[1] John D. Anderson. *Computational Fluid Dynamics: The Basics with Applications*. McGraw-Hill, New York, NY, 1995.

[2] Randall J. LeVeque. *Finite Volume Methods for Hyperbolic Problems*. Cambridge University Press, Cambridge, UK, 2002.